Dineth Wijesinghe

DevOps Engineer Intern

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SUMMARY

Information Technology undergraduate at the University of Moratuwa with hands-on project experience building AWS infrastructure, containerized microservices, serverless workflows, and observability systems. Skilled in Terraform, Kubernetes (EKS/K3s), Docker, GitHub Actions, Linux, and core AWS services. Built repeatable delivery pipelines, event-driven services, health checks, failure-handling workflows, and OpenTelemetry monitoring; seeking a DevOps or Cloud Engineering internship.

EDUCATION

BSc (Hons) in Information Technology **2024 – Present**

University of Moratuwa, Moratuwa, Sri Lanka

- **GPA :- 3.6 / 4.0**

G.C.E. Advanced Level **2020 – 2022**

H/North Kanumuldeniya National School, Hambantota, Sri Lanka

- Combined Mathematics | Physics | ICT · **Z-Score :- 1.7991**

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, SQL, Bash

Backend Development: Node.js, Express.js, Bun, Hono, FastAPI, RESTful APIs, WebSockets

Frontend Development: React, React 18, Vite, Tailwind CSS, HTML5

Databases & ORM: PostgreSQL, MySQL, SQLite, MongoDB, Redis, DynamoDB, Entity Framework Core

Cloud & DevOps: AWS (EKS, EC2, ALB, RDS, Lambda, S3), Docker, Kubernetes (K3s), Terraform, Jenkins, GitHub Actions, CI/CD, Linux, Nginx, Git

KEY PROJECTS

CloudSight – Cloud Infrastructure Observability Platform (Group Project) GitHub Repo

▶ **Technologies :-** Node.js, Express, React, Python, scikit-learn, Docker, AWS (EC2, RDS, ECR, Amplify), PostgreSQL, Redis, Nginx, GitHub Actions

- Implemented a GitHub Actions delivery workflow that builds SHA-tagged backend and ML Docker images, pushes them to Amazon ECR, and deploys the services to EC2 with Docker Compose behind Nginx.
- Integrated an asynchronous six-detector anomaly pipeline combining hard thresholds, rolling baselines, trend analysis, Isolation Forest, log-spike detection, and stale-telemetry checks with Redis-backed caching.
- Containerized the backend, ML worker, and Redis services with restart policies, dependency ordering, and persistent volumes; documented deployment and anomaly runbooks for operational handoff.

AWS K3s OpenTelemetry Observability Lab GitHub Repo

▶ **Technologies :-** AWS EC2, K3s Kubernetes, OpenTelemetry Collector, Honeycomb, OTLP/HTTPS, Docker, Linux, Bash

- Provisioned a K3s cluster on AWS EC2 for a 30+ microservice lab with Security Group isolation and namespace-level CPU and memory quotas.
- Applied Kubernetes liveness and readiness probes with resource controls to support automatic recovery and predictable scheduling under failure.
- Configured an OpenTelemetry Collector pipeline to export telemetry over OTLP/HTTPS to Honeycomb for p95/p99 trace analysis and bottleneck investigation.

ShopSphere – Containerized Microservices Platform on AWS GitHub Repo

▶ **Technologies :-** Amazon EKS, Kubernetes, AWS ALB Ingress Controller, Terraform, Docker, Multi-AZ VPC, RDS PostgreSQL, DynamoDB, SNS, SQS, HPA

- Provisioned a live Amazon EKS microservices platform with modular Terraform across Multi-AZ public, private, and database subnets, ECR repositories, and private worker nodes.
- Configured AWS Load Balancer Controller path-based routing, Metrics Server-backed HPA, resource controls, and non-root service containers for controlled scaling and safer ingress.
- Decoupled checkout, inventory, payment, and notification flows with SNS/SQS fan-out and Dead-Letter Queues, then added pod-kill, traffic-spike, and poison-pill chaos tests.

AWS Serverless & Event-Driven Order Processing Platform GitHub Repo

▶ **Technologies :-** AWS Lambda, Amazon API Gateway, Amazon SNS, Amazon SQS (DLQ), DynamoDB, CloudWatch, AWS X-Ray, Terraform, GitHub Actions (OIDC)

- Built a Terraform-managed serverless order workflow using API Gateway, Lambda, SNS fan-out, SQS worker queues, DynamoDB, and CloudWatch monitoring.
- Implemented poison-pill isolation with SQS Dead-Letter Queues, partial-batch failure handling, and DynamoDB conditional-write idempotency for exactly-once business effects.
- Automated formatting, linting, unit-test coverage, Terraform validation, and OIDC-authenticated deployment through GitHub Actions with X-Ray tracing and operational runbooks.

AWS Three-Tier Scalable Web Application Infrastructure GitHub Repo

▶ **Technologies :-** AWS (EC2, ALB, Auto Scaling Groups, RDS PostgreSQL, CloudWatch), Terraform, Node.js, Express, React 18, Docker, Pino, Nginx, GitHub Actions (OIDC)

- Provisioned a Terraform-managed three-tier AWS environment with public ALB subnets, private EC2 Auto Scaling application subnets, isolated database subnets, and custom health checks (`/health/ready`).
- Configured ALB and ASG health checks, CPU-based scaling, CloudWatch dashboards and alarms, CloudWatch Agent logs, and GitHub Actions OIDC for deployment.
- Added Docker Compose staging, structured Pino JSON logging, encrypted SSM parameters, and configurable RDS PostgreSQL backups; Multi-AZ is available as a Terraform option.

UniNet – Cross-Platform Network Automation Utility GitHub Repo

▶ **Technologies :-** Bash, PowerShell, Linux NetworkManager Dispatcher, macOS launchd, Windows Task Scheduler, curl, GitHub Actions

- Engineered a no-third-party-runtime-dependency utility that authenticates university captive portals across Linux, macOS, and Windows for Cisco/NEC, Aruba, and Ruijie networks.
- Implemented event-driven NetworkManager, launchd, and Windows Task Scheduler hooks that trigger only on Wi-Fi state changes without a continuously running background daemon.
- Built cross-platform install and release automation with dependency healing, shell and PowerShell validation, simulation tests, and GitHub Releases.

Tunnix – Self-Hosted Developer Tunneling & Ingress Platform GitHub Repo

▶ **Technologies :-** Go, gorilla/websocket, Bun, Docker, AWS (EC2, ECS, ECR, EBS, CloudFront, Amplify, S3, SSM), Nginx, Let's Encrypt, SQLite (WAL), TypeScript, Bash

- Developed a Go WebSocket ingress gateway that multiplexes HTTP traffic between public clients and cross-platform CLI agents using correlation IDs and bidirectional frames.
- Containerized the Bun/Hono control plane and Go gateway, provisioning EC2/ECS, ECR, EBS-backed SQLite WAL storage, SSM parameters, and Nginx TLS ingress with Terraform.
- Automated GitHub Actions delivery for the client, server, gateway, agent binaries, and infrastructure validation, including CloudFront/Amplify delivery and S3 versioned artifact publishing.

CERTIFICATIONS & COURSES

- AWS Cloud Essentials – Amazon Web Services · IAM, VPC, EC2, RDS, and Cloud Architecture *(Ongoing)*
- LFS158: Introduction to Kubernetes – The Linux Foundation · Architecture, Deployments & Pod Networking *(Ongoing)*
- LFS101: Introduction to Linux – The Linux Foundation · Linux Administration, Shell Scripting & System Processes
- CCNA (Cisco Certified Network Associate) – Enterprise Networking, Routing & Switching, TCP/IP *(Ongoing)*

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Batch Representative** | **Faculty of Information Technology, University of Moratuwa** 2026 – Present
- **Programme Committee Member** | **Riseup Mora 2026, University of Moratuwa** 2026
- **Senior Prefect** | **H/North Kanumuldeniya National School** 2019 – 2020

REFERENCES

Mr. Indika Wimalasuriya
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